



Essay topic: T cell exhaustion

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Abstract

T cell exhaustion limits the efficacy of immunotherapies in solid tumors, and tumor-infiltrating myeloid cells (TIMs) are key extrinsic cellular drivers of this dysfunctional state. This essay examines specifically how tumor-associated macrophages and myeloid-derived suppressor cells actively shape T cell exhaustion in the tumor microenvironment (TME). We then critically evaluate current therapeutic approaches aiming to either prevent TIM recruitment or reprogram them to anti-tumor phenotypes to aid reinvigoration of terminally exhausted T cells. Lastly, we briefly discuss major barriers in current TIM-targeting therapies and suggest future directions of TIM-targeting strategies to overcome T cell exhaustion-driven resistance to existing immunotherapies.

List of Abbreviations

ATAC-seq: assay for transposase-accessible chromatin with high-throughput sequencing

APC: antigen-presenting cells

CAF: cancer-associated fibroblast

ccRCC: clear cell renal cell carcinoma

CCR: C-C chemokine receptor

CEBP β : CCAAT/enhancer-binding protein beta

ChARs: chromatin accessible regions

ChIP-seq: chromatin immunoprecipitation followed by sequencing

CSF: colony-stimulating factor

CTLA-4: cytotoxic T lymphocyte-associated protein 4

CXCR: CXC chemokine receptor

Dnmt3a: DNA methyltransferase 3A

GCN2: general control nonderepressible 2

ICB: immune checkpoint blockade

IFN: interferon

IL: interleukin

IR: inhibitory receptor

IRF8: interferon-regulatory factor 8

ITAM: immunoreceptor tyrosine-based activation motif

LAG-3: lymphocyte-activation gene 3

LCMV: lymphocytic choriomeningitis virus

MDSC: myeloid-derived suppressor cell

MHC: major histocompatibility complex

mTORC1: mammalian target of rapamycin complex 1

NFAT: nuclear factor of activated T-cells

NF- κ B: nuclear factor kappa-light-chain-enhancer of activated B cells

NR4A: nuclear receptor subfamily 4A

PDAC: pancreatic ductal adenocarcinoma

PD-1: programmed cell death protein-1

PI3K γ : phosphatidylinositol 3-kinase gamma

SHP-2: Src homology-2 protein tyrosine phosphatase

SIRP α : signal-regulatory protein alpha

S6K: S6 kinase

TAM: tumor-associated macrophages

TAN: tumor-associated neutrophils

TADC: tumor-associated dendritic cells

Tex: exhausted T cells

TCF-1: T cell factor 1

TCR: T cell receptor

TDLN: tumor draining lymph nodes

TGF: transforming growth factor

TIGIT: T cell immunoreceptor with Ig and ITIM domains

TIM: tumor-infiltrating myeloid cells

TIM3: T cell immunoglobulin and mucin domain-containing protein 3

TME: tumor microenvironment

TNF- α : tumor necrosis factor α

TOX: thymocyte selection-associated high mobility group box protein

1. Introduction

T lymphocytes are central to adaptive immunity, mediating cellular responses against intracellular pathogens and malignant cells. Upon T cell receptor (TCR) recognition of the peptide major histocompatibility complex (MHC) complex on antigen-presenting cells (APCs) (signal 1), co-stimulatory interactions (signal 2), and cytokine signaling (signal 3), naïve CD4⁺ or CD8⁺ T cells become activated, which later differentiate into effector subsets that execute cytotoxic functions and regulate immune response. Following antigen clearance and resolution of inflammation, most activated T cells undergo apoptosis, while a small subset differentiates into long-lived memory T cells. However, during persistent exposure to antigen and inflammatory signals, which is common in chronic viral infections and tumor microenvironments (TME), this differentiation process becomes dysregulated. Under such conditions, T cells enter a hypofunctional state known as T cell exhaustion (Tex), characterized by progressive loss of effector function such as cytokine production, sustained upregulation of inhibitory receptors (IRs), and transcriptional and epigenetic reprogramming ([1](#), [2](#)). Exhausted T cells have been identified across a wide range of malignancies, including hepatocellular carcinoma, melanoma, and head and neck cancer ([3](#), [4](#)), where their presence is frequently linked to poor clinical outcomes. The rationale of immune checkpoint blockade (ICB) therapies is to block the IRs to restore the anti-tumor activity of pre-existing T cells ([5](#)). However, terminal T cell exhaustion in many cancer types has been one of the major barriers to immune checkpoint blockade (ICB) therapies, as these terminal exhausted T cells cannot be functionally reinvigorated ([6](#)). Consequently, only about 12.5% of patients eligible for ICB therapy in the US are predicted to respond, underscoring the importance of understanding the mechanism driving terminal T cell exhaustion ([7](#)).

Tumor-infiltrating myeloid cells (TIMs), which include tumor-associated macrophages (TAMs), myeloid-derived suppressor cells (MDSCs), tumor-associated neutrophils (TANs), and tumor-associated dendritic cells (TADCs), function as a double-edged sword in the TME. Taking TAMs as an example, these cells can exert anti-tumor functions by presenting antigens to activate adaptive immune cells and by secreting pro-inflammatory cytokines that induce vascular damage and tumor necrosis. However, TAMs can also acquire a pro-tumor, M2-like phenotype, which promotes angiogenesis and secretion of immunosuppressive cytokines that inhibit tumor-

infiltrating lymphocytes (TILs) (8-10). TAM abundance is closely linked to T cell exhaustion and poor prognosis, with TAMs forming long-lasting immunological synapses with T cells to induce and sustain their exhaustion state (11). Furthermore, TAMs, instead of tumor cells, are found to be the primary source of persistent antigen presentation driving terminal T cell exhaustion (4). TAMs exemplify how myeloid populations within the TME shape T cell exhaustion. This phenomenon extends beyond TAMs, as other TIM subsets likewise contribute to driving and sustaining T cell exhaustion.

The hallmarks of T cell exhaustion are well established, yet the extrinsic cellular factors that sustain this dysfunctional state in TME remain incompletely understood. Emerging evidence indicates that TIMs are not passive bystanders but active participants in promoting and maintaining T cell exhaustion through multiple immunosuppressive mechanisms. However, the underlying molecular mechanisms and specific interactions within different TMEs are still not fully delineated. This essay examines how TIMs, particularly TAMs, drive T cell exhaustion and critically evaluates therapeutic strategies aimed at targeting these populations to restore T cell function and overcome immunotherapy resistance.

2. Overview and defining features of T cell exhaustion

T cell exhaustion was first described by Zajac *et al.* in lymphocytic choriomeningitis virus (LCMV) infected mouse models ([12](#)). Fundamental work conducted over the following decade using the LCMV mouse model established that persistent antigen exposure and TCR engagement are primary drivers of T cell exhaustion ([1](#), [12](#), [13](#)). Chronic viral infection models provided a reproducible model for investigating fundamental principles of exhaustion. However, early studies of exhaustion in tumor models lagged behind because of the TME's heterogeneity and complexity. In cancer, chronic antigen exposure is often compounded by metabolic restriction, hypoxia, and constant exposure to immunosuppressive cytokines and metabolites released by TIMs. These extrinsic cues regulate the activity of intrinsic transcriptional regulators such as NFAT, NR4A, and TOX to enforce exhaustion both transcriptionally and epigenetically, shaping the differentiation trajectory and irreversibility of T cell exhaustion. This section mainly examines the hallmarks of exhaustion and the intrinsic mechanisms that promote T cell exhaustion.

2.1. Hallmarks of T cell exhaustion

As exhaustion progresses, CD8⁺ T cells gradually lose cytolytic activity and the ability to produce key inflammatory cytokines, initially IL-2, followed by TNF- α and IFN- γ . In addition, exhausted T cells exhibit markedly reduced proliferative capacity, altered metabolism, and fail to expand effectively in response to antigenic stimulation and homeostatic cytokines ([12](#), [14](#)). A defining feature of this dysfunctional state is the sustained upregulation of inhibitory receptors, with programmed cell death protein-1 (PD-1) being the most prominent. PD-1 is transiently induced upon TCR engagement, which acts as a 'brake' that negatively regulates T cell activation. Mechanistically, upon binding with its ligand PD-L1 or PD-L2, PD-1 recruits phosphatase SHP-2, which dephosphorylates tyrosine residues on ITAMs of CD3 ζ and CD28 cytoplasmic tail. This disrupts both TCR and CD28 costimulatory signaling ([15](#)), leading to dampened T cell activation and proliferation. Under acute conditions, PD-1 expression is downregulated once antigen is cleared. In environments where chronic antigen exposure is present, however, exhausted CD8⁺ T cells maintain sustained and progressively increasing levels of PD-1 over time. Although blocking PD-1 signaling in Tex can partially reverse their dysfunctional state and enhance T cell activity ([16](#), [17](#)),

the reversal is only partial as the underlying transcriptional and epigenetic profile remains largely intact. These findings suggest that PD-1 is a central negative regulator of T cell activation and provide mechanistic evidence that sustained PD-1 signaling is a critical marker of T cell exhaustion. Other inhibitory receptors, such as CTLA-4, TIGIT, TIM3, and LAG3, also contribute to the progression of exhaustion. All hallmarks of exhaustion are summarized in [Figure 1A \(18\)](#).

2.2. Transcriptional and epigenetic reprogramming of Tex

Persistent antigen stimulation reshapes the transcriptional and epigenetic landscape of T cells, locking them into a distinct dysfunctional state. NFAT family proteins, key transcription factors activated downstream of TCR signaling, promote exhaustion when activated in the absence of their canonical binding partner AP-1. Among them, NFAT5 has recently been identified to be uniquely upregulated in tumor-infiltrating CD8⁺ T cells to drive exhaustion but not in chronic viral infections ([19](#)). Using an engineered, constitutively active form of NFAT incapable of interacting with AP-1, it was revealed that it can bind directly to regulatory regions of many exhaustion-associated genes. Specifically, unpartnered NFAT induces transcription of *Pdcd1* (PD-1), as well as the master regulator of exhaustion *Tox* and the *Nr4a* family genes (*Nr4a1*, *Nr4a2*, *Nr4a3*) ([20-22](#)).

Multiple studies identified that NR4A induces the expression of inhibitory receptors and represses genes involved in effector function, thus accelerating exhaustion ([21](#), [23](#)). Consistent with this, deletion of *Nr4a1/2* enhances the accumulation and function of TCF-1⁺ stem-like Tex precursors, suggesting that NR4A factors restrict progenitor expansion under chronic stimulation and promote terminal differentiation ([24](#)). However, recent ChIP-seq analysis in CD8⁺ T cells suggests that the NR4A1 is highly expressed in TCF-1⁺ progenitor exhausted cells, where it sustains their quiescent, self-renewing state by upregulating *Tcf7* while restraining effector differentiation ([25](#)). This apparent discrepancy may reflect stage-specific or compensatory effects among NR4A family members, where transient NR4A1 activity supports progenitor stability, while sustained or redundant NR4A signaling enforces terminal exhaustion. Downstream effectors regulated by NR4As at different stages of exhaustion require further investigation. Other essential transcriptional regulators of exhaustion are illustrated in [Figure 1B](#).

The epigenetic landscape of Tex is distinct from that of effector (Teff) and memory T cells (Tmem). For genome-wide analysis of Tex, ATAC-seq was performed to map the chromatin accessible regions (ChARs) in the genomes of Tex, Teff, and Tmem. Sen *et al.* found that 44.48% of all ChARs differ between Tex and Teff/Tmem, revealing that exhaustion is accompanied by extensive chromatin remodeling and represents a discrete differentiation lineage. As an illustrative example, they examined the PD-1 locus and identified an exhaustion-specific enhancer region upstream of *Pdcd1* that remains accessible only in Tex. Deletion of this enhancer using CRISPR/Cas9 reduced PD1 expression (26). Additionally, DNA demethylation was observed at the PD1 locus in Tex, which switches on PD1 expression, suggesting a causal relationship between histone modification and exhaustion. Further validation using *Dnmt3a* (DNA methyltransferase 3A) conditional knockout (cKO) revealed that loss of DNMT3A retains effector functions and stemness, as effector and stemness-associated loci such as *Ifng* and *Tcf7* remain demethylated (27). Whether these “epigenetic scars” of terminally exhausted T cells can be dynamically remodeled in vivo remains uncertain, which limits the potential for full functional restoration after ICB. Therefore, characterizing subsets of Tex and identifying the immunotherapy-responding population is critical.

2.3. Heterogeneity and developmental trajectory of Tex

Exhausted T cells can be broadly categorized based on their distinct functional, epigenetic, and transcriptional characteristics into progenitor-exhausted (Tex_prog), intermediate Tex, and terminal-exhausted (Tex_term) subsets, which arise along a linear differentiation trajectory (28). Utzschneider *et al.* first identified a population of TCF1⁺PD-1⁺ memory-like CD8⁺ T cells in mice LCMV model, which is characterized by the expression of TCF1 (*Tcf7*), a critical transcription factor. Although the TCF1⁺ cells exhibit traits of exhaustion, they retain proliferative potential and act as Tex_prog that sustain terminally exhausted populations and maintain immune responses during chronic infection (29). Consistent results were later observed in mouse tumor models by Siddiqui *et al.*, who demonstrated that TCF1⁺PD-1⁺ stem-like tumor-infiltrating CD8⁺ T cells possess proliferative potential in response to therapeutic vaccination and PD-1 blockade. These cells are capable of self-renewal and can differentiate into TCF1⁺PD-1⁺ terminally exhausted cells, sustaining the intratumoral Tex pool (30).

To better characterize the two distinct subsets, Miller *et al.* performed single-cell RNA sequencing on CD8⁺ T cells from mice chronically infected with LCMV, identifying Tex_prog as Slamf6⁺Tim-3⁻ and Tex_term as Slamf6⁻Tim-3⁺. To further examine the lineage relationship, each population was transferred into congenically marked, tumor-bearing mice. Tex_prog cells were able to give rise to both Slamf6⁺Tim-3⁻ and Slamf6⁻Tim-3⁺ cells, whereas only Tim-3⁺ cells were recovered for Tex_term transfer, confirming a unidirectional linear differentiation trajectory (28). Between these two states, a transitory effector-like CD101⁺Tim3⁺CX3CR1⁺ intermediate state with pro-inflammatory cytokine and granzyme B expression derived from Slamf6⁺Tim-3⁻ progenitors was characterized. This intermediate subset exhibits strong cytotoxic ability and represents the main effector arm of the exhausted lineage (31, 32).

Comparable differentiation trajectories have been observed in human cancers. Tumor-specific CD8⁺ T cells are initially primed in tumor-draining lymph nodes (TDLNs), where they acquire a Tex_prog phenotype marked by TCF-1 expression and proliferative capacity. These cells subsequently migrate into the TME, where they progress to terminal differentiation (18, 33). Consistent with findings from murine models, the proliferating TCF-1⁺CD8⁺ T cells within TDLNs serve as a reservoir that continually replenishes the exhausted T cell pool in TME, enabling transient cytotoxic activity (34). Extending this concept, Huang *et al.* identified that the bona fide responders to PD-1/PD-L1 blockade are not intratumoral Tex but a group of TCF-1⁺TOX⁻ tumor-specific memory CD8⁺ T cells residing in the TDLNs. This group of cells serves as the upstream reservoir of TDLN Tex_prog. Additionally, local administration of PD-L1 inhibitors into TDLNs effectively controls tumor growth, whereas surgical removal of TDLNs abolishes the anti-tumor effect of ICB (35).

In summary, T cell exhaustion progresses along a unidirectional differentiation trajectory, characterized by the expression of various cell surface markers and different functional capacities, within a continuous spectrum rather than discrete stages (Figure 1C). The existence and plasticity of intermediate subsets with residue effector potential highlight a therapeutic window of reversibility before terminal epigenetic fixation occurs. However, the extent to which tumor-infiltrating myeloid cells shape each stage of this trajectory remains a critical and unresolved question for understanding how to therapeutically reverse exhaustion within the TME.

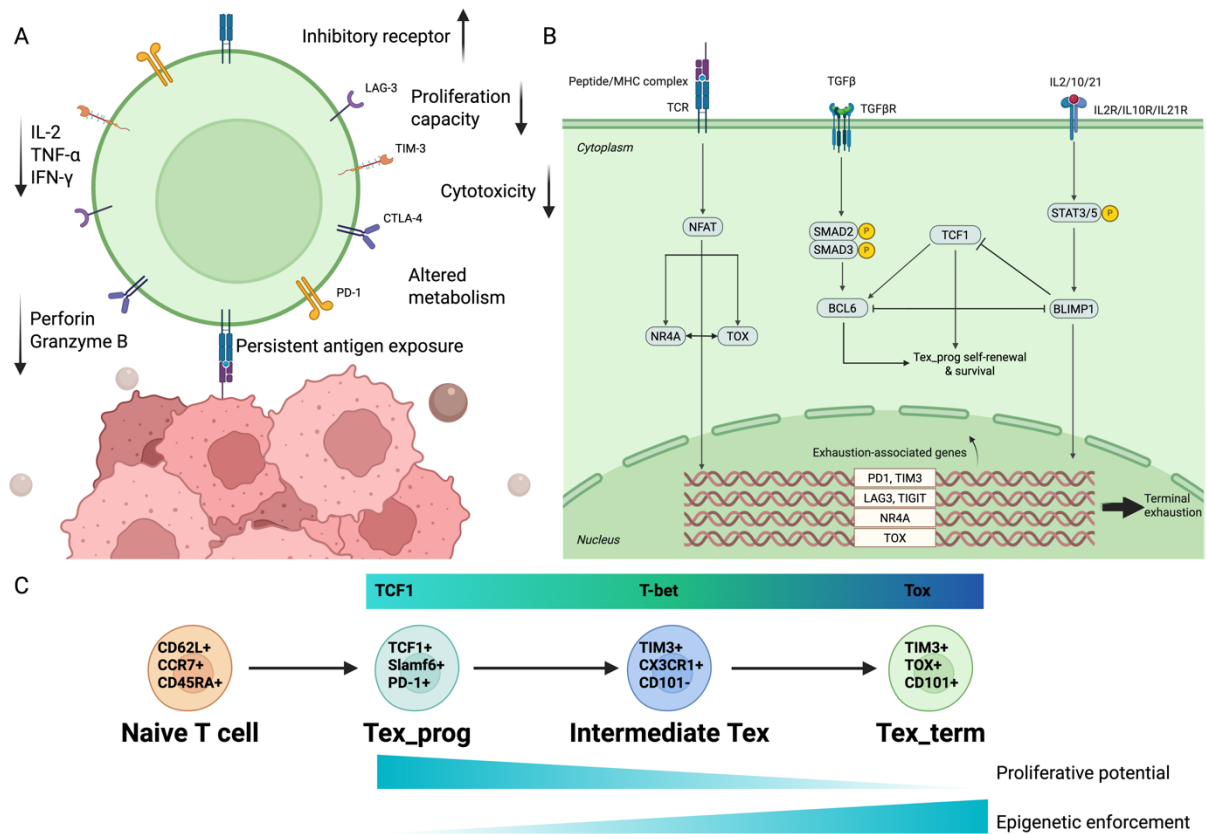


Figure 1 Defining features of T cell exhaustion and intrinsic regulators (Created in BioRender). (A) Hallmarks of T cell exhaustion. (B) Transcriptional regulators of T cell exhaustion: TCR-NFAT-TOX/NR4A signaling establishes and sustains the exhausted state, while progression is driven by a TCF1/BCL6-BLIMP1 antagonistic axis. TCF1 is a critical transcription factor maintaining progenitor survival and self-renewal, while BCL6 inhibits further differentiation and preserves progenitor identity. IL2-STAT5 and IL10-STAT3 induce BLIMP1, which antagonizes TCF1 and BCL6 signaling, promoting effector-like and terminal Tex differentiation. (C) Differentiation trajectory of T cell exhaustion with characteristic markers illustrated.

3. Role of TIMs in shaping T cell exhaustion

TME is a highly heterogeneous environment containing multiple types of TIMs, cancer-associated fibroblasts (CAFs), and cancer cells. Interactions between tumor-infiltrating T cells and TIMs, either directly through chronic TCR-antigen recognition and inhibitory receptor engagement, or indirectly through the release of immunosuppressive factors, create an immunosuppressive niche that drives T-cell exhaustion. General mechanisms of TIMs driving exhaustion are illustrated in detail in [Figure 2](#).

3.1. Tumor-associated macrophages (TAMs)

Tumor-associated macrophages are the most abundant population of TIMs within the TME and play crucial roles in both initiating and sustaining T cell exhaustion. In murine melanoma models, Kersten *et al.* revealed a spatialtemporal co-dependency between TAMs and Tex. They identified a bidirectional TAM-T cell feedback loop in which exhausted CD8⁺ T cells secrete colony-stimulating factor 1 (CSF1) and chemokines to recruit and program TAMs toward an antigen-presenting phenotype that sustains chronic interactions. These TAMs, despite expressing normal antigen-presentation and co-stimulatory machinery, form a weak stimulatory, long-lasting synapse that primes T cells for exhaustion. Additionally, TAMs and Tex are colocalized at highly hypoxic and poorly vascularized regions, which further exacerbates exhaustion ([11](#)).

TAMs express high levels of interferon-regulatory factor 8 (IRF8), a transcription factor essential for TAM and cDC1 differentiation and expansion. RNA-seq analysis revealed that IRF8-sufficient TAMs, compared with IRF8-deficient counterparts, are enriched for genes involved in antigen presentation, exhaustion-driving molecule (SLAMF7), and T cell recruiting chemokines, while showing only minimal expression of the DC-specific transcription factor *Zbtb46*. This indicates that TAMs acquire a DC-like, antigen-presenting phenotype in an IRF8-dependent manner while maintaining their macrophage identity. Functionally, IRF8-sufficient TAMs were capable of cross-presenting tumor antigens and inducing PD-1 expression on CD8⁺ T cells. In contrast, macrophage-specific deletion of IRF8 impaired TAM proliferation and antigen-presentation capacity, reducing PD-1⁺ exhausted T cells while enhancing granzyme B⁺ effector T cells and tumor control. These suggest that IRF8-dependent antigen presentation is critical for sustaining exhausted T cell states within the tumor. Consistent with these findings, TAMs in clear cell renal cell carcinoma (ccRCC)

express high levels of IRF8, and an IRF8-TAM gene signature correlates with T cell exhaustion and poor patient survival in highly infiltrated tumors (36). However, the study did not detect impaired expression of canonical cross-presentation genes in IRF8-deficient TAMs, suggesting that these cells may primarily acquire peptide-MHC complexes from tumor cells through a “cross-dressing” mechanism like their DC counterparts rather than relying on endogenous antigen processing. This can be further investigated by knocking out cross-presentation-related genes and examining the effect of KO on driving T cell exhaustion.

Furthermore, Polania *et al.* demonstrated that TAMs, rather than tumor cells, are the primary source of chronic antigen exposure driving terminal differentiation of Tex within the TME. In glioblastoma CCR2^{-/-} mice, treated with an anti-CSF1R antibody, which creates a TAM-deficient TME, a significantly reduced number of Tex_{term} were observed. The frequency of Tex_{term} is also positively correlated with the persisting MHC1⁺ on TAMs. Supporting this, in an in vitro co-culture system, peptide-pulsed TAMs induced higher PD-1⁺TIM-3⁺ expression and greater transition of Tex_{prog} to Tex_{term} than equally pulsed tumor cells (4). This study overturns the long-held assumption that tumor cells are the main source of chronic antigen exposure, positioning TAMs as the central APC maintaining persisting TCR engagement that enforces exhaustion in solid tumors.

The new model of TAM-driven exhaustion reveals a spatial and temporal interdependence between macrophages and CD8⁺ T cells. However, it presents a paradox. IRF8 expression is necessary for both TAMs and TADCs to acquire antigen-presentation ability, but the same program transforms them into chronic stimulators. This pushes Tex_{prog} toward terminal exhaustion by sustaining a low-strength TCR engagement within hypoxic niches rather than activating its effector function. Still, it is unclear whether IRF8⁺ TAMs are essential for causing exhaustion or if they are a result of an already suppressive environment. This raises the question of whether targeting this pathway can reverse exhaustion without disrupting important antigen presentation.

3.2. Myeloid-derived suppressor cells (MDSCs)

Myeloid-derived suppressor cells (MDSCs) are a diverse population of aberrantly activated, immature myeloid cells produced during tumor-driven myelopoiesis, where they become major enforcers of the immunosuppressive TME (37). They are broadly

categorized into monocytic MDSCs (M-MDSCs) and granulocytic/polymorphonuclear MDSCs (PMN-MDSCs), each deploying distinct suppressive programs once within the tumor. In the TME, M-MDSCs produce high levels of nitric oxide, IL-10, TGF- β , and upregulate PD-L1, whereas PMN-MDSCs generate reactive oxygen species, peroxynitrite, arginase-1, and prostaglandin E2 (PGE2). Among these, arginase 1 depletes L-arginine, which is a critical metabolite for T cell activation and proliferation (37). Collectively, these mediators establish a chronic inhibitory environment that, together with persistent antigen stimulation, drives T cell exhaustion. Consistent with this, preventing CXCR2-mediated MDSC trafficking to TME significantly enhances the efficacy of anti-PD1 therapy (38). However, this study used CXCR2^{-/-} hematopoietic chimeras, meaning that all bone-marrow-derived cells lacked CXCR2. As neutrophils and other myeloid subsets also depend on CXCR2 for trafficking, the observed increased efficacy may not be attributed exclusively to MDSC depletion, and contributions from other CXCR2-dependent populations should be considered.

A key intrinsic regulator of MDSC-mediated immunosuppression is general control nonderepressible 2 (GCN2), a stress-responsive Ser-Thr kinase that senses nutrient limitations and regulates transcription and translation of multiple genes involved in immunosuppression and exhaustion promotion. Deletion of GCN2 in the myeloid lineage produces a markedly more inflamed TME, with M-MDSCs increasing expression of co-stimulatory CD86 and MHC I/II, and PMN-MDSCs downregulating PD-L1, CD206, and CD39. Consistent with reduced MDSC suppressive activity, GCN2-deficient mice exhibit significantly decreased expression of exhaustion markers PD-1, LAG-3, TIGIT, and TIM-3 on exhausted intratumoral CD8⁺ T cells (39). However, whether GCN2 deletion produces only an acute functional rescue or genuinely reverses the exhaustion-associated chromatin state remains unknown. TCR lineage tracing coupled with single-cell RNA and ATAC-seq before and after myeloid-specific GCN2 deletion is needed to determine whether Tex_term clones lose exhaustion-associated chromatin accessibility and reacquire Tex_prog features, thereby validating GCN2 as a therapeutic target.

4. Targeting TIMs to reinvigorate exhausted T cells

4.1. Preventing TIMs recruitment, proliferation, and survival

TIM abundance in the TME is often related to poor prognosis and a high proportion of terminally exhausted T cells. Therefore, targeting TIM infiltration, proliferation, and survival is a feasible strategy. The CCL2-CCR2 axis is critical for TAM/MDSC recruitment, angiogenesis, and tumor metastasis. Theoretically, blocking this pathway can reduce exhaustion and enhance tumor control. However, clinical trials using either anti-CCL2 monoclonal antibodies (Carlumab) or CCR2 small molecule inhibitors (PF-4136309) targeting TAMs have been ineffective as monotherapy or in combination, thus terminated ([8](#)).

Another targetable pathway is the CSF1-CSF1R axis, which governs the migration, differentiation, and survival of TAMs/MDSCs and their progenitors. Preclinical data in mice and humans have shown that anti-CSF1 antibody (RG7155) leads to CSF-1-dependent TAM depletion and shifts the immune balance to CD8⁺ T cell activity ([40](#)). However, in the proneural glioblastoma model, CSF-1R inhibition was found to block glioma progression by reducing the polarization of M2-like, tumor-promoting macrophages rather than depleting them, with their continued survival aided by glioma-secreted GM-CSF and IFN- γ ([41](#)). CSF-1 signaling inhibition causes divergent TAM outcomes because TAM reliance on CSF-1 is highly TME-dependent. This is confirmed by the clinical success of CSF-1R blockade in diffuse-type tenosynovial giant cell tumor, a non-malignant tumor characterized by CSF-1 overexpression ([8](#)). However, most malignant tumors show minimal response to CSF-1 inhibition alone.

Preventing TIM recruitment or survival often leads to compensatory expansion of other TIM subsets that maintain the immunosuppressive environment, sustaining T cell exhaustion. The myeloid redundancy, together with the heterogeneity across different kinds of tumors, explains why standalone treatments targeting TIMs rarely translate into long-lasting clinical benefits. Therefore, identifying dominant immunosuppressive subsets of myeloid cells in each specific tumor type, supported by detailed molecular profiles, is essential for predicting how patients will respond and for designing combination therapies with chemotherapy or ICB that can avoid myeloid redundancy and more effectively reinvigorate exhausted T cells.

4.2. Reprogramming TIMs to anti-tumor phenotypes and block immunosuppressive effector functions

The high plasticity of TIMs allows the possibility to functionally reprogram them by altering their intrinsic signaling or changing the external environmental cues involved in polarization. Phosphatidylinositol 3-kinase gamma (PI3K γ) activation drives M2-like TAMs polarization. Its downstream signaling promotes the transcription of immunosuppressive factors, such as arginase 1 (Arg 1), via activation of the mTORC1/S6K/CEBP β axis, while simultaneously dampening NF- κ B-dependent proinflammatory cytokine production ([42](#)). In pancreatic ductal adenocarcinoma (PDAC) mouse models, PI3K γ blockade reprograms TAMs toward an IL-12⁺/iNOS⁺ inflammatory M1-like state and restores a degree of CD8⁺ T cell cytotoxicity, hence partially rescues exhaustion and enhances tumor control ([43](#), [44](#)). The small molecule PI3K γ -selective inhibitor, eganelisib, has been developed and progressed through multiple Phase II trials. In metastatic triple-negative breast cancer, eganelisib in combination with chemotherapy and anti-PD1 therapy has shown the highest efficacy. This combination has produced clear evidence of TAM reprogramming, extracellular-matrix remodeling, and enhanced T cell activation ([42](#), [45](#)). However, the dependency of eganelisib on combination regimens marks its limited standalone potency. Also, it raises the concern whether this can translate into long-term survival benefits in larger, randomized phase III trials in solid tumors. The compensatory effect of redundant M2 TAM polarization pathways after PI3K γ inhibition should be investigated. Besides, the current strategy of directly blocking PI3K γ raises concerns about off-target effects on healthy cells, given the importance of the PI3K pathway in somatic cells. Thus, identifying tumor-specific downstream effectors of the PI3K γ pathway that could be targeted with greater precision could be the next step in the field to reduce side effects.

The CD47/SIRP α axis is a key 'Don't Eat Me' signal as CD47 is ubiquitously expressed on normal cells and binds SIRP α on macrophages and dendritic cells to prevent phagocytosis. Tumor cells exploit this by overexpressing CD47, while TAMs maintain high SIRP α levels, suppressing their phagocytic activity and reinforcing an immunosuppressive, M2-like phenotype. Antagonizing this axis not only restores phagocytosis but also activates TADCs and repolarizes TAMs to an anti-tumor phenotype. CD47/SIRP α blockade increases antigen uptake, enhances MHC-I presentation, improves co-stimulatory signaling, and reduces the secretion of

immunosuppressive cytokines by TAMs, thereby enhancing CD8⁺ T cell priming and slowing down exhaustion (44, 46). Clinically, targeting the CD47/SIRPα axis has shown efficacy in hematological malignancies, but its benefit in solid tumors remains uncertain. Several ongoing phase II clinical trials of anti-CD47/SIRPα (NCT04313881, NCT04751383, NCT04675294, NCT04675333) targeting head and neck squamous cell carcinoma and colorectal cancer are used in combination with other targeted therapies (8). As CD47 is ubiquitously expressed, minimizing anti-CD47 off-target binding to healthy cells, particularly red blood cells, remains a major obstacle for safe systemic delivery. At present, CD47/SIRPα blockade is best viewed as an adjunctive strategy whose success will depend on rational combinations. Although targeting this pathway is mechanistically feasible, it remains uncertain whether the degree of TAM and TADC reprogramming achieved by CD47/SIRPα inhibition is sufficient to counteract the profoundly suppressive and antigen-rich TME that continually drives CD8⁺ T cells toward terminal exhaustion. Other potential druggable pathways are illustrated in detail in [Figure 2](#).

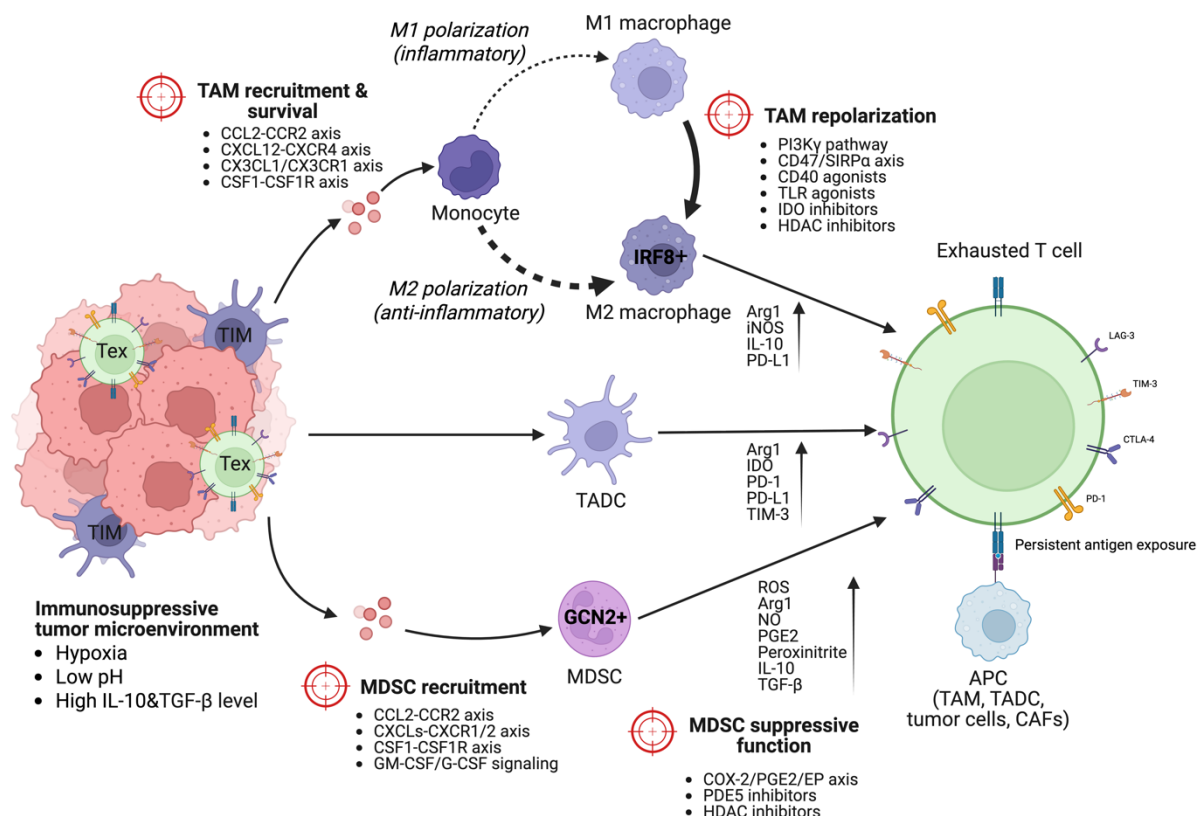


Figure 2 TIMs' role in driving and sustaining exhaustion with key targetable pathways indicated (Created with BioRender). 1) **Immunosuppressive molecules: Arginase 1 (Arg1)** – an enzyme that depletes extracellular L-arginine, a critical amino acid for T cell signaling, proliferation, and metabolism; **NO (nitric oxide)**, a reactive nitrogen species that inhibits T cell activation and induces apoptosis; **iNOS** (induced nitric oxide synthase) - an

enzyme that synthesizes NO; **IL-10&TGFβ** - immunosuppressive cytokines; **IDO** (indoleamine 2,3-dioxygenase) – depletes tryptophan, a critical amino acid for T cell survival, by converting it to kynurenine metabolites; **ROS** (reactive oxygen species) – inhibits TCR signaling, promotes MDSC expansion, and drives M2 TAM polarization; **PGE2** (Prostaglandin E₂) – suppresses NK cells and T cells; **Peroxynitrite** – a type of reactive nitrogen species. **2) Undiscussed targetable pathways:** **CD40 agonist** - activates APCs to drive strong effector immune responses.; **TLR agonist** - triggers innate pattern recognition pathways to awaken suppressed antitumor immunity; **IDO inhibitors** - block tryptophan catabolism to prevent kynurenine-driven T cell suppression; **HDAC inhibitors** – alter chromatin accessibility to reprogram suppressive immune cells; **COX-2/PGE2/EP axis** – inhibition of this axis releases suppression on T cells and NK cells; **PDE5 inhibitors** – PDE5 inhibition increases cGMP, downregulating Arg1 and NOS.

5. Conclusion and Discussion

T cell exhaustion progresses from Tex_{prog} through an effector-like stage to terminally exhausted populations in the TME. In this process, transcriptional regulators such as the NFAT-TOX-NR4A and the TCF1-BCL6-BLIMP1 axis, together with increased chromatin accessibility at inhibitory receptor loci, lead to an epigenetically fixed, irreversible Tex_{term} state. One major barrier to TIM-targeting therapy is precisely this irreversibility. Once terminal exhaustion occurs, the mentioned intrinsic transcriptional and epigenetic programs primarily maintain this state. Meanwhile, external immunosuppressive signals from TIMs strengthen this identity instead of maintaining it. Consequently, strategies aimed at preventing TIMs recruitment and survival (CCL2-CCR2 inhibitors, CSF1-CSF1R inhibitors) or reprogramming TIMs (PI3K γ inhibition, CD47/SIRP α antagonism) can only enhance effector activity of intermediate and progenitor Tex and improve responsiveness to ICBs, but cannot reprogram Tex_{term} back to progenitor-like states. The second major barrier is the heterogeneity and redundancy of the TIMs population. Targeting one suppressive pathway often triggers compensatory upregulation of alternative exhaustion-driving pathways. Besides, the heterogeneity of TIMs also highlights the need to identify shared components across different TIM subsets, as targeting these is more likely to generate broad and long-lasting therapeutic effects. Moreover, the field's predominant focus on CD8⁺ exhaustion overlooks emerging evidence that CD4⁺ T cell exhaustion, characterized by reduced IL-21 production, impaired TAM/TADC licensing, and defective Tfh-like, indirectly drives CD8⁺ exhaustion ([47](#)). Thus, the interactions between TIMs and CD4⁺ T cells warrant further investigation and may offer novel therapeutic opportunities.

In my view, the most promising therapeutic direction involves combinatorial strategies that reprogram, rather than deplete, multiple dominant myeloid populations in combination with ICB or chemotherapy. Besides, the success of this approach relies on a deeper mechanistic understanding of how myeloid cells, tumor cells, and TILs interact to generate an immunosuppressive TME that shapes T cell exhaustion trajectories.

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